lxNavbar
lx-
Header
lx-
Footer

Hyperconvexity and domains of discontinuity for surface groups in $SL_3\mathbb{C}$

BY

Jennifer Vaccaro
BS, Olin College of Engineering, 2017
MS, University of Illinois at Chicago, 2022

THESIS [PhD candidates may use DISSERTATION here instead]

Submitted as partial fulfillment of the requirements
for the degree of Doctorate of Philosophy in Mathematics
in the Graduate College of the
University of Illinois Chicago, 2026

Chicago, Illinois

Defense Committee:
Emily Dumas
Name
Name
Name, Outside Members Need Affiliation Here

Accessibility Statement

An accessible EPUB version of this document is available at [URL] or can be obtained by contacting [contact details].

Dedication

For my grandmother Martha, who also saw the joy in geometry.

Acknowledgements

Emily (integrity, knowledge, high standards)
Sam (love, support, meals, no pressure)
Family (you don't know why I wanted to do this, but that's ok)
Friends (kept me company, kept me happy)
Other math teachers and mentors
Gertrude (tiniest companion)

Preface

Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Curabitur pretium tincidunt lacus. Nulla gravida orci a odio. Nullam varius, turpis et commodo pharetra, est eros bibendum elit, nec luctus magna felis sollicitudin mauris.

Integer in mauris eu nibh euismod gravida. Duis ac tellus et risus vulputate vehicula. Donec lobortis risus a elit. Etiam tempor. Ut ullamcorper, ligula eu tempor congue, eros est euismod turpis, id tincidunt sapien risus a quam.

Contents

Accessibility Statement ii

Dedication iii

Acknowledgements iv

Preface vi

List of Figures ix

List of Tables x

List of Abbreviations xi

Summary xiii

Chapter 1. Introduction 1
1. Motivation 2

Chapter 2. Background 4
1. Group theory 5

Chapter 3. Main Result 6
1. Statements 7
2. A couple proofs 7
3. Action of $\rho(g)$ on F^3 8
4. Extend to generic Richardson varieties of Λ 9
5. F matches or switches Richardson varieties 10

Chapter 4. Another result 12
1. Second main result 13

Chapter 5. Vitae 14

Appendix A. Technical Lemmas 16

Bibliography 18

List of Figures

1 This is a torus 2
2 The Snake Lemma 2

List of Tables

1 Example Table 3

List of Abbreviations

- $\mathbb{R}$ The real numbers
- $\mathbb{C}$ The complex numbers

Summary

Cras mollis scelerisque nunc. Nullam arcu. Aliquam consequat. Curabitur augue lorem, dapibus quis, laoreet et, pretium ac, nisi. Aenean magna nisl, mollis quis, molestie eu, feugiat in, orci. In hac habitasse platea dictumst.

Maecenas fermentum consequat mi. Donec fermentum. Pellentesque malesuada nulla a mi. Duis sapien sem, aliquet nec, commodo eget, consequat quis, neque. Aliquam faucibus, elit ut dictum aliquet, felis nisl adipiscing sapien, sed malesuada diam lacus eget erat.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Vivamus luctus urna sed urna ultricies ac tempor dui sagittis. In condimentum facilisis porta.

Sed nec diam eu diam mattis viverra. Nulla fringilla, orci ac euismod semper, magna diam porttitor mauris, quis sollicitudin sapien justo in libero.

Fusce lacinia quam nisi, euismod ultrices nulla malesuada ut. Curabitur sit amet magna quam. Praesent in libero vel turpis pellentesque egestas sit amet vel nunc.

Morbi lectus risus, iaculis vel, suscipit quis, luctus non, massa. Fusce ac turpis quis ligula lacinia aliquet. Mauris ipsum. Nulla metus metus, ullamcorper vel, tincidunt sed, euismod in, nibh.

Quisque volutpat condimentum velit. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos himenaeos. Nam nec ante. Sed lacinia, urna non tincidunt mattis, tortor neque adipiscing diam, a cursus ipsum ante quis turpis.

Cras elementum ultrices diam. Maecenas ligula massa, varius a, semper congue, euismod non, mi. Proin porttitor, orci nec nonummy molestie, enim est eleifend mi, non fermentum diam nisl sit amet erat.

Duis semper. Duis arcu massa, scelerisque vitae, consequat in, pretium a, enim. Pellentesque congue. Ut in risus volutpat libero pharetra tempor. Cras vestibulum bibendum augue.

Praesent egestas leo in pede. Praesent blandit odio eu enim. Pellentesque sed dui ut augue blandit sodales. Vestibulum ante ipsum primis in faucibus orci luctus et ultrices posuere cubilia Curae.

Aliquam nibh. Mauris ac mauris sed pede pellentesque fermentum. Maecenas adipiscing ante non diam sodales hendrerit.

Ut velit mauris, egestas sed, gravida nec, ornare ut, mi. Aenean ut orci vel massa suscipit pulvinar. Nulla sollicitudin. Fusce varius, ligula non tempus aliquam, nunc turpis ullamcorper nibh, in tempus sapien eros vitae ligula.

Pellentesque rhoncus nunc et augue. Integer id felis. Curabitur aliquet pellentesque diam. Integer quis metus vitae elit lobortis egestas.

Lorem ipsum dolor sit amet, consectetuer adipiscing elit. Morbi vel erat non mauris convallis vehicula. Nulla et sapien. Integer tortor tellus, aliquam faucibus, convallis id, congue eu, quam.

Mauris ullamcorper felis vitae erat. Proin feugiat, augue non elementum posuere, metus purus iaculis lectus, et tristique ligula justo vitae magna.

Aliquam convallis sollicitudin purus. Praesent aliquam, enim at fermentum mollis, ligula massa adipiscing nisl, ac euismod nibh nisl eu lectus.

Fusce vulputate sem at sapien. Vivamus leo. Aliquam euismod libero eu enim. Nulla nec felis sed leo placerat imperdiet. Aenean suscipit nulla in justo.

Suspendisse cursus rutrum augue. Nulla tincidunt tincidunt mi. Curabitur iaculis, lorem vel rhoncus faucibus, felis magna fermentum augue, et ultricies lacus lorem varius purus.

Sed adipiscing ornare risus. Morbi est est, blandit sit amet, sagittis vel, euismod vel, velit. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

Nullam nonummy metus. Vestibulum volutpat pretium libero. Cras id dui. Aenean ut eros et nisl sagittis vestibulum.

Nulla facilisi. Integer lacinia sollicitudin massa. Cras metus. Sed aliquet risus a tortor. Integer id quam.

CHAPTER 1

Introduction

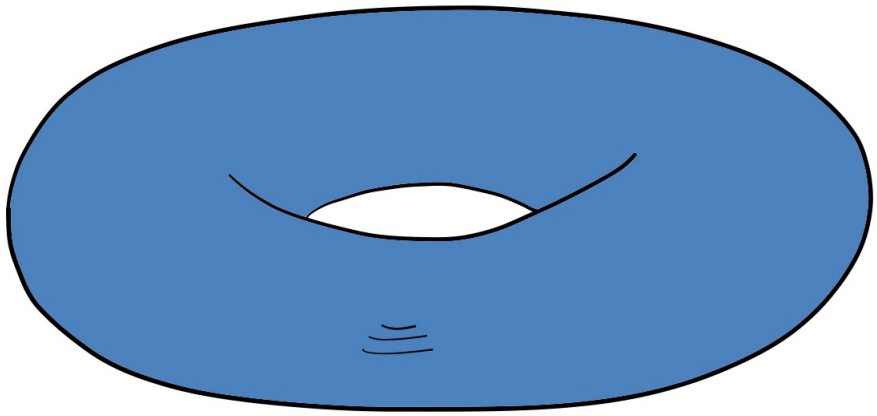

FIGURE 1. This is a torus

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & A' \arrow[r, "f'"] \arrow[d, "a"'] & B' \arrow[r, "g'"] \arrow[d, "b"'] & C' \arrow[r] \arrow[d, "c"'] & 0 \\
0 \arrow[r] & A \arrow[r, "f"] \arrow[d, "a''"'] & B \arrow[r, "g"] \arrow[d, "b''"'] & C \arrow[r] \arrow[d, "c''"'] & 0 \\
0 \arrow[r] & A'' \arrow[r, "f''"] \arrow[d] & B'' \arrow[r, "g''"] \arrow[d] & C'' \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}$$

FIGURE 2. The Snake Lemma

This is the introduction chapter. We cite some classic works [**1, 2**].

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Morbi tincidunt nisi sed neque volutpat, et imperdiet metus congue. Donec interdum, neque at varius vehicula, elit odio tempus sapien, sed finibus lorem felis nec turpis. Aliquam erat volutpat. Mauris vehicula imperdiet erat, non aliquet magna commodo id.

Sed porttitor urna ac ipsum rhoncus, vel dictum risus commodo. Integer ultrices dui sit amet pulvinar tempor. Phasellus nec tellus eu turpis sodales accumsan. Proin ultr

THEOREM 1.1. *This is a theorem*

We reference Theorem 1.1.

University of Illinois Chicago.

1. Motivation

1.1. Historical context. A brief overview of how the problem developed.

$$\int_0^1 f(x)dx = 2 \tag{1}$$

How to solve (1)

1.2. Open questions. Some questions remain open for future work.
Note: I have not tested the accessibility of this table.

Monkeys	Lions
100	200

Table 1. Example Table

CHAPTER 2

Background

This chapter gives necessary background.

1. Group theory

A group is a set G with a binary operation satisfying closure, associativity, identity, and inverses.

THEOREM 2.1. *Every finite subgroup of the multiplicative group of a field is cyclic.*

PROOF. This is a standard result from algebra. $\square$

CHAPTER 3

Main Result

Let Γ the fundamental group of a closed hyperbolic surface. Let $\rho, \rho' : \Gamma \to SL_3\mathbb{C}$ Anosov. We use the following notation related to ρ and ρ'.

- $\xi, \xi' : \partial\Gamma \to \mathcal{F}^3$ refers to the continuous, transverse, ρ- or ρ'- equivariant, dynamics-preserving boundary maps.
- $\Lambda, \Lambda' \subset \mathcal{F}^3$ refers to the limit curves, which are the images of ξ, ξ'.
- $Th(\Lambda), Th(\Lambda')$ refers to the thickened limit curve.
- $Th(x), Th(x')$ refer to the thickening of specific flags $x \in \Lambda, x' \in \Lambda'$.

1. Statements

We call ρ, ρ' *topologically conjugate* if there exists a homeomorphism $F : \mathcal{F}^3 \to \mathcal{F}^3$ such that for all $g \in \Gamma$ and all $x \in \mathcal{F}^3$, then $F(\rho(g) \cdot x) = \rho'(g) \cdot F(x)$.

THEOREM 3.1. *Hyperconvexity is closed under topological conjugacy. In other words, if ρ and ρ' are topologically conjugate, then ρ is hyperconvex if and only if ρ' is hyperconvex.*

The proof of this theorem uses two lemmas, in which we assume $F : \mathcal{F}^3 \to \mathcal{F}^3$ is a topological conjugacy between ρ and ρ'. Clearly $F(\Lambda) = \Lambda'$. This mapping is pointwise, i.e. for all $\alpha \in \partial\Gamma$, then $F(\xi(\alpha)) = \xi'(\alpha)$. So for $x \in \Lambda$, we can use the notation $x' := F(x) \in \Lambda'$.

For any pair of transverse flags $x, y \in \mathcal{F}^3$, consider the irreducible 0-dimensional Richardson varieties $R^0(x, y)$.

EXAMPLE 3.1. *Suppose x and y are the attracting and repelling flags of $\rho(g) = diag(\lambda, \mu, \nu)$ where $|\lambda| > |\mu| > |\nu|$. Then $R^0(x, y)$ consists of $\{x, x_y, x^y, y_x, y^x, y\}$ where:*

- $x = \{x\} \cap \alpha_{121}(y) = (e_1, e_1 + e_2)$
- $x_y = \alpha_1(x) \cap \alpha_{12}(y) = (e_1, e_1 + e_3)$
- $x^y = \alpha_2(x) \cap \alpha_{21}(y) = (e_2, e_1 + e_2)$
- $y_x = \alpha_{12}(x) \cap \alpha_1(y) = (e_3, e_1 + e_3)$
- $y^x = \alpha_{21}(x) \cap \alpha_2(y) = (e_2, e_2 + e_3)$
- $y = \alpha_{121}(x) \cap \{y\} = (e_3, e_2 + e_3)$.

Our topological conjugacy F preserves the 0-dimensional Richardson varieties almost pointwise.

LEMMA 3.1.1. *For all x,y $\in \Lambda^{(2)}$,$Fsends R^0_{x,y}$ to $R^0_{x',y'}$.*

- *If F* matches *Richardson varieties, then $F(x_y) = x'_{y'}$, $F(x^y) = x'^{y'}$.*
- *If F* switches *Richardson varieties, then $F(x_y) = x'^{y'}$, $F(x^y) = x'_{y'}$.*

As a consequence of the argument in Lemma 3.1.1, we also have the following result:

COROLLARY 3.1.1. *For all $x \in \Lambda$, then $F(Th(x)) = Th(x')$.*

The relative position of 0-dimensional Richardson varieties determines hyperconvexity.

LEMMA 3.1.2. *The representation ρ is hyperconvex if and only if for all $x, y, z \in \Lambda^{(3)}$, then $R^0_{x,y} \cap R^0_{x,z} = \{x\}$.*

2. A couple proofs

We first prove our main theorem, given our supporting lemmas.

PROOF. (Theorem 3.1) Suppose $F : \mathcal{F}^3 \to \mathcal{F}^3$ is a topological conjugacy between ρ and ρ'.

We begin with the forward direction. Assume that ρ is hyperconvex. Let $x, y, z \in \Lambda^{(3)}$. By Lemma 3.1.2, then $x_y \neq x_z$ and $x^y \neq x^z$. By Lemma 3.1.1, then $\{F(x_y), F(x^y), F(x_z), F(x^z)\} = \{x'_{y'}, x'^{y'}, x'_{z'}, x'^{z'}\}$ are distinct. Since x, y, z were general, then this holds for all triples in Λ and ρ' is hyperconvex.

The other direction proceeds similarly. Assume that ρ is *not* hyperconvex. By Lemma 3.1.2, there exist $x, y, z \in \Lambda^{(3)}$ such that $x_y = x_z$ or $x^y = x^z$. By Lemma 3.1.1, then F either matches or switches Richardson varieties. So one of the following is true: $x'_{y'} = x'_{z'}$ or $x'^{y'} = x'^{z'}$. Therefore ρ' is *not* hyperconvex. □

We also prove our lemma linking Richardson varieties to hyperconvexity.

PROOF. (Lemma 3.1.2) Suppose ρ is Anosov. Let $x = (x_1, x_2), y = (y_1, y_2), z = (z_1, z_2) \in \Lambda^{(3)}$. Since x, y, z are pairwise transverse, then $y_x, y^x, z_x, z^x \notin R^0_{x,y} \cap R^0_{x,z}$. It suffices to check the pairs $x_y, x_z \in \alpha_1(x)$ and $x^y, x^z \in \alpha_2(x)$.

By definition, $x_y = (x_1, x_1 + y_1)$ and $x_z = (x_1, x_1 + z_1)$, which are distinct if and only if $x_1 + y_1 \neq x_1 + z_1$, or equivalently $z_1 \not\subset x_1 + y_1$.

Similarly, $x^y = (x_2 \cap y_2, x_2)$ and $x^z = (x_2 \cap z_2, x_2)$, which are distinct if and only if $x_2 \cap y_2 \neq x_2 \cap z_2$, or equivalently $x_2 \cap y_2 \not\subset z_2$.

So $R^0_{x,y} \cap R^0_{x,z} = \{x\}$ for all $x, y, z \in \Lambda^{(3)}$ is equivalent to ρ hyperconvex. □

The remainder of this chapter is devoted to the proof of Lemma 3.1.1.

3. Action of $\rho(g)$ on F^3

We describe the dynamics of $\rho(g) \cdot \mathcal{F}^3$, where $g \in \Gamma$ infinite order. Since ρ is Anosov, then we know $\rho(g)$ is strongly loxodromic, i.e. its eigenvalues have distinct magnitude. That means it has an attracting and repelling fixed boundary points $g^+, g^- \in \partial\Gamma$. Let $x = \xi(g^+), y = \xi(g^-)$.

Under $\rho(g)$, most of $\mathcal{F}^3$ is attracted by x and repelled by y. We define the subset of $\mathcal{F}^3$ attracted to x as the "uphill" $U_{\rho(g)}(x) := \{f \in F^3 : \rho(g^n) \cdot f \to x\}$. Similarly, we define the subset of $\mathcal{F}^3$ repelled by y as the "downhill" $D_{\rho(g)}(y) := \{f \in \mathcal{F}^3 : \rho(g^{-n}) \cdot f \to y\}$. Set $E_{\rho(g)} = \mathcal{F}^3 \setminus (U_{\rho(g)}(x) \cap D_{\rho(g)}(y))$.

The set $E_{\rho(g)}$ is nonempty, since it contains x and y, and closed, since it is complement of an open set. It also contains the other fixed points of $\rho(g) \cdot \mathcal{F}^3$, which are precisely $R^0_{x,y}$. (This is straightforward to check using Example 3.1, which is generic because up to conjugation $\rho(g) = diag(\lambda, \mu, \nu)$.)

We can describe subsets of $E_{\rho(g)}$ explicitly by their attracting and repelling fixed point. These subsets are the irreducible 1-dimensional Richardson varieties $R^1_{x,y}$:

- $span(x, x_y) = \alpha_1(x) = \alpha_1(x) \cap \alpha_{121}(y) = \overline{U(x) \cap D(x_y)}$
- $span(x, x^y) = \alpha_2(x) = \alpha_2(x) \cap \alpha_{121}(y) = \overline{U(x) \cap D(x^y)}$
- $span(x_y, y_x) = \alpha_{12}(x) \cap \alpha_{12}(y) = \overline{U(x_y) \cap D(y_x)}$
- $span(x^y, y^x) = \alpha_{21}(x) \cap \alpha_{21}(y) = \overline{U(x^y) \cap D(y^x)}$
- $span(y_x, y) = \alpha_1(y) = \alpha_{121}(x) \cap \alpha_1(y) = \overline{U(y_x) \cap D(y)}$
- $span(y^x, y) = \alpha_2(y) = \alpha_{121}(x) \cap \alpha_2(y) = \overline{U(y^x) \cap D(y)}$

These varieties are not disjoint, e.g. $\{x\} = span(x, x_y) \cap span(x, x^y)$, but the points of intersection are exactly the 0-dimensional Richardson varieties. So $E_{\rho(g)}$ is homeomorphic to the union of six copies of $\mathbb{CP}^1$, each identified with two neighbors in a cyclic order $(x, x_y, y_x, y, y^x, x^y)$.

For the following results, let $F : \mathcal{F}^3 \to \mathcal{F}^3$ be a topological conjugacy between $\rho, \rho' : \Gamma \to SL_3\mathbb{C}$. We use the notation $x' = F(x), y' = F(y)$.

Proposition 3.1. *Let $g \in \Gamma$, and suppose $x = \xi(g^+), y = \xi(g^-)$. Then $F(x_y) \in \{x'_{y'}, x'^{y'}\}$.*

Proof. Let $S : S^1 \to E_{\rho(g)}$ continuous and injective such that it passes through $(x, x_y, y_x, y, y^x, x^y)$ in cyclic order. Without loss of generality, we may assume $S(0) = x$, $S(\pi/3) = x_y$, etc. Since F is a topological conjugacy, then it must preserve the exceptional set, i.e. $F(E_{\rho(g)}) = E_{\rho'(g)}$.

Then $F \cdot S : S^1 \to E_{\rho'(g)}$ is also injective. We also know $F \cdot S(0) = x'$ and $F \cdot S(\pi) = y'$. So by injectivity and the cyclic order of $R^0_{x',y'}$ in $E_{\rho'(g)}$, then either $F \cdot S(\pi/3) = x'_{y'}$ or $F \cdot S(\pi/3) = x'^{y'}$. The remaining points $F \cdot S(n\pi/3)$ where $n \in \{2, 4, 5\}$ are determined by this choice as well. □

4. Extend to generic Richardson varieties of Λ

In this section, we still assume $F : \mathcal{F}^3 \to \mathcal{F}^3$ is a topological conjugacy between ρ, ρ'. We extend our argument from fixed points of infinite order words in Γ to generic Richardson varieties in Λ.

First, we extend Proposition 3.1 to Richardson varieties of the form x_z where x is the attracting fixed point of an infinite order element $g \in \Gamma$ and z is a generic point on Λ.

Proposition 3.2. *Let $g \in \Gamma$ infinite order, and define $x := \xi(g^+)$. Let $z \in \Lambda \setminus \{x\}$. Then $F(x_z) \in \{x'_{z'}, x'^{z'}\}$.*

We use the following result from group theory:

Remark 3.1. *Let Γ hyperbolic with boundary $\partial\Gamma \equiv S^1$, and let $x, z \in \partial\Gamma$. Then there exists a quasigeodesic sequence $\{g_i\}_{i=1} \in \Gamma^{\mathbb{N}}$ such that for all $y \in \partial\Gamma$:*

$$lim_{i\to\infty} g_i \cdot y = \begin{cases} x : y = x \\ z : y \neq x \end{cases} . \tag{2}$$

Proof. (Proposition 3.2) If $z = \xi(g^-)$ then we are done, so assume not. Set $y = \xi(g^-)$.

Let $\{g_i\}_{i=1} \in \Gamma^{\mathbb{N}}$ as in Rem. 3.1, such that $\lim_{i=1} \rho(g_i) \cdot y = z$ and $lim_{i=1}\rho(g_i) \cdot x = x$. Let $x_i := \rho(g_i) \cdot x$ and $y_i := \rho(g_i) \cdot y$, and let $x'_i := F(x_i), y'_i := F(y_i)$.

By Prop. **??** then for all $i \in \mathbb{N}$ we have $F(x_{iy_i}) \in \{x'_{iy'_i}, x'^{y'_i}_i\}$. Clearly $\lim_{i\to\infty} x_{iy_i} = x_z$, and $\lim_{i\to\infty} x'_i = x'$ and $\lim_{i\to\infty} y'_i = z'$. By continuity of F and compactness of $\mathcal{F}^3$, then

$$F(x_z) = F(\lim_{i\to\infty} x_{iy_i}) \tag{4}$$

$$= \lim_{i\to\infty} F(x_{iy_i}) \tag{5}$$

$$\in \{\lim_{i\to\infty} x'_{iy'_i}, \lim_{i\to\infty} x'^{y'_i}_i\} \tag{6}$$

$$= \{x'_{z'}, x'^{z'}\}. \tag{7}$$

□

Next, we extend Proposition 3.2 to Richardson varieties of the form w_z where both w and z are generic points of Λ.

PROPOSITION 3.3. *For all $w, z \in \partial\Gamma^{(2)}$, then $F(w_z) \in \{w'_{z'}, w'^{z'}\}$.*

PROOF. Let $w, z \in \Lambda$. If w is the attracting fixed point of an infinite order element, we are done by Prop 3.2, so assume not.

Let $\{x_i\}_{i=1} \in \Lambda^{\mathbb{N}}$ such that each $x_i = \xi(g_i)$ the attracting fixed point of an infinite order word, and $\lim_{i\to\infty} x_i = w$. This sequence exists because attracting fixed points of $\rho(g)$ where g infinite order are dense in Λ. Define $x'_i = F(x_i)$. By continuity of F, we have

$$F(w) = F(\lim_{i\to\infty} x_i) \tag{8}$$
$$= \lim_{i\to\infty} F(x_i) \tag{9}$$
$$= \lim_{i\to\infty} (x'_i) \tag{10}$$
$$= w'. \tag{11}$$

Then by Prop 3.2 and by continuity of F, we have

$$F(w_z) = F(\lim_{i\to\infty} x_{iz}) \tag{12}$$
$$= \lim_{i\to\infty} F(x_{iz}) \tag{13}$$
$$\in \{\lim_{i\to\infty} (x'_{iz'}), \lim_{i\to\infty} (x'^{z'}_i)\} \tag{14}$$
$$= \{w'_{z'}, w'^{z'}\}. \tag{15}$$

□

5. F matches or switches Richardson varieties

In this section, we highlight how F preserves the balanced ideal thickening centered at each point $x \in \Lambda$. Observe that $Th(x) = \alpha_1(x) \cup \alpha_2(x) = span(x, x_y) \cup span(x, x^y)$ for some $y \in \Lambda \setminus \{x\}$.

PROOF. (Corollary 3.1.1) By Prop. 3.1, $\{F(x_y), F(x), F(x^y)\} = \{x'_{y'}, x, x'^{y'}\}$. Observe that F is a homeomorphism that preserves dynamics in that $F(U_{\rho(g)}) = U_{\rho'(g)}$ and $F(D_{\rho(g)}) = D_{\rho'(g)}$. So then

$$F(Th(x)) = F(span(x, x_y) \cup span(x, x^y)) \tag{16}$$
$$= span(x', x'_{y'}) \cup span(x', x'^{y'}) \tag{17}$$
$$= Th(x'). \tag{18}$$

□

We also show that F either *matches* or *switches* Richardson varieties, meaning that it either sends preserves α_1 for each $x \in \Lambda$, or swaps it with α_2.

COROLLARY 3.1.2. *If for some $w \in \Lambda$ we have $F(\alpha_1(w)) = \alpha_1(w')$, then for all $z \in \Lambda$ we have $F(\alpha_1(z)) = \alpha_1(z')$. That is, F either* matches *or* switches *Richardson varieties.*

PROOF. Suppose for some $w \in \Lambda$ we have $F(\alpha_1(w)) = \alpha_1(w')$. Let $z \in \Lambda \ \{w\}$. Since F is a homeomorphism, by Cor. 3.1.1 we have $F(\alpha_1(z)) \in \{\alpha_1(z'), \alpha_2(z')\}$.

For contradiction, suppose that $F(\alpha_1(z)) = \alpha_2(z')$. Then for all $g \in \Gamma$, by the definition of a topological conjugacy, we have $F(\alpha_1(\rho(g) \cdot z)) = \alpha_2(\rho'(g) \cdot z')$.

Since $\rho(g) \cdot z$ is dense in Λ, then there exists a sequence $\{g_i\}_{i=1} \in \Gamma^{\mathbb{N}}$ such that $\lim_{i\to\infty} \rho(g_i) \cdot z = w$. Define $z_i = \rho(g_i) \cdot z$.

Then

$$\begin{aligned}
\alpha_1(w') &= F(\alpha_1(w)) && (19)\\
&= F(\alpha_1(\lim_{i\to\infty} z_i)) && (20)\\
&= \lim_{i\to\infty} F(\alpha_1(z_i)) && (21)\\
&= \lim_{i\to\infty} \alpha_2(w) && (22)\\
&= \alpha_2(w). && (23)
\end{aligned}$$

This is a contradiction. □

Clearly $x_y \in \alpha_1(x)$, so if F matches Richardson varieties then $F(x_y) = x'_{y'}$, and if F switches Richardson varities then $F(x_y) = x'^{y'}$.

So for all $x, y \in \Lambda$, a topological conjugacy F sends $R^0_{x,y}$ to $R^0_{x',y'}$, and either matches or switches the irreducible 0- and 1-dimensional Richardson varieties. This completes the proof of Lemma 3.1.1.

CHAPTER 4

Another result

1. Second main result

Another significant theorem.

CHAPTER 5

Vitae

For UIC this can be a full CV or a short version (e.g. half a page) with previoius degrees, work experience and publications.

APPENDIX A

Technical Lemmas

Here we collect some supporting lemmas.

Bibliography

[1] R. Hartshorne, *Algebraic Geometry*, Springer-Verlag, New York, 1977.

[2] D. Mumford, *Abelian Varieties*, Oxford University Press, 1970.

[3] J. Draisma, E. Horobet, G. Ottaviani, B. Sturmfels, and R. R. Thomas, "The Euclidean distance degree of an algebraic variety," *arXiv:1309.0049* (2013). Available at: https://arxiv.org/abs/1309.0049